MTH129 Introduction to AI for Mathematical Sciences (MS)

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SMP AI Team

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Linear Regression and Least Square

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Regression Model

Regression model:

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1d} \\ 1 & x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nd} \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_d \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

Component-wise Form

For each observation $i = 1, \ldots, n$:

$$y_i = w_0 + w_1 x_{i1} + w_2 x_{i2} + \cdots + w_d x_{id} + \varepsilon_i$$

- w_0: Intercept
- x_{ij}: j-th feature of i-th observation
- ε_i: Noises

Problem Setting in Matrix/Vector Form

Regression model:

$$\mathbf{y} = \mathbf{X}\mathbf{w} + \varepsilon$$

Matrix Dimensions

- $\mathbf{y} \in \mathbb{R}^{n\times 1}$: Response vector ($n$ observations)
- $\mathbf{X} \in \mathbb{R}^{n\times(d+1)}$: Design matrix ($d$ features, including bias)
- $\mathbf{w} \in \mathbb{R}^{(d+1)\times 1}$: Weight vector (parameters to estimate)
- $\varepsilon \in \mathbb{R}^{n\times 1}$: Noise vector

Assumption on Noise distribution

Assumption on Noise distribution: Multivariate Gaussian

$$\varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$$

which implies:

$$\mathbf{y}|\mathbf{X}, \mathbf{w} \sim \mathcal{N}(\mathbf{X}\mathbf{w}, \sigma^2 \mathbf{I})$$

Probability Density Function

$$p(\mathbf{y}|\mathbf{X}, \mathbf{w}, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\mathbf{w})^T(\mathbf{y} - \mathbf{X}\mathbf{w})\right)$$

- $\sigma^2\mathbf{I}$: Diagonal covariance (independent homoscedastic errors)
- $\mathbf{I}$: $n \times n$ identity matrix

Important

Here, we are assuming all y_i share the independent and identical errors, which is almost impossible in any cases, this is called homoscedastic, we will introduce more general case later (heteroscedastic).

Least Squares Method

The general idea of the least squares method is to find the minimized sum of square errors (minimize the noise).

Objective Function

$$J(\mathbf{w}) = \frac{1}{2}\sum_{i=1}^{n}(y_i - \mathbf{w}^T\mathbf{x}_i)^2$$

Least Squares Method

Expanding the Objective Function

$$
\begin{aligned}
J(\mathbf{w}) &= \frac{1}{2}(\mathbf{y} - \mathbf{X}\mathbf{w})^T(\mathbf{y} - \mathbf{X}\mathbf{w}) \\
&= \frac{1}{2}[\mathbf{y}^T\mathbf{y} - \mathbf{y}^T\mathbf{X}\mathbf{w} - (\mathbf{X}\mathbf{w})^T\mathbf{y} + (\mathbf{X}\mathbf{w})^T\mathbf{X}\mathbf{w}] \\
&= \frac{1}{2}[\mathbf{y}^T\mathbf{y} - 2\mathbf{w}^T\mathbf{X}^T\mathbf{y} + \mathbf{w}^T\mathbf{X}^T\mathbf{X}\mathbf{w}]
\end{aligned}
$$

Note: $\mathbf{y}^T\mathbf{X}\mathbf{w} = (\mathbf{X}\mathbf{w})^T\mathbf{y}$ since both are scalars

Gradient Calculation

Take derivative w.r.t. $\mathbf{w}$:

$$\begin{aligned}\nabla_{\mathbf{w}} J(\mathbf{w}) &= \nabla_{\mathbf{w}} \left(\frac{1}{2}\mathbf{y}^T\mathbf{y} - \mathbf{w}^T\mathbf{X}^T\mathbf{y} + \frac{1}{2}\mathbf{w}^T\mathbf{X}^T\mathbf{X}\mathbf{w} \right) \\ &= -\mathbf{X}^T\mathbf{y} + \frac{1}{2} \cdot 2\mathbf{X}^T\mathbf{X}\mathbf{w} \\ &= \mathbf{X}^T\mathbf{X}\mathbf{w} - \mathbf{X}^T\mathbf{y}\end{aligned}$$

Key matrix derivative identities

$$\frac{\partial}{\partial \mathbf{w}}(\mathbf{w}^T\mathbf{a}) = \mathbf{a} \qquad \frac{\partial}{\partial \mathbf{w}}(\mathbf{w}^T\mathbf{A}\mathbf{w}) = 2\mathbf{A}\mathbf{w} \qquad \text{(for symmetric } \mathbf{A}\text{)}$$

Closed-form Solution

Set gradient to zero:

$$\mathbf{X}^T\mathbf{X}\mathbf{w} - \mathbf{X}^T\mathbf{y} = \mathbf{0}$$
$$\mathbf{X}^T\mathbf{X}\mathbf{w} = \mathbf{X}^T\mathbf{y}$$

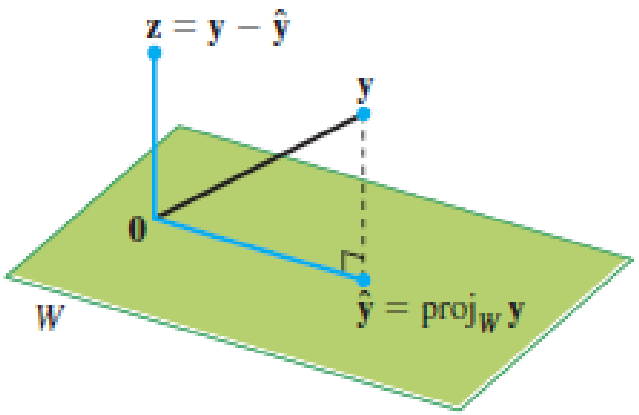
The Normal Equations:

$$\boxed{\mathbf{w}^* = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}}$$

- $\mathbf{X}^T\mathbf{X}$ must be invertible $\Leftrightarrow$ $\mathbf{X}$ has full column rank. (Why? When not full rank?)
- If singular, use pseudo-inverse: $\mathbf{w}^* = \mathbf{X}^\dagger\mathbf{y}$
- Geometric interpretation: Projection in feature space
- Limitations: Non-invertible $\mathbf{X}^T\mathbf{X}$, overfitting

Geometric Interpretation

$$\mathbf{X}^T(\mathbf{y} - \mathbf{X}\mathbf{w}^*) = \mathbf{0}$$
$$\mathbf{X}^T\mathbf{r}^* = \mathbf{0}$$



Residual vector $\mathbf{r}^*$ is orthogonal to:

- All columns of $\mathbf{X}$
- The column space $\mathcal{C}(\mathbf{X})$

Solution $\mathbf{X}\mathbf{w}^*$ is the orthogonal projection of $\mathbf{y}$ onto $\mathcal{C}(\mathbf{X})$

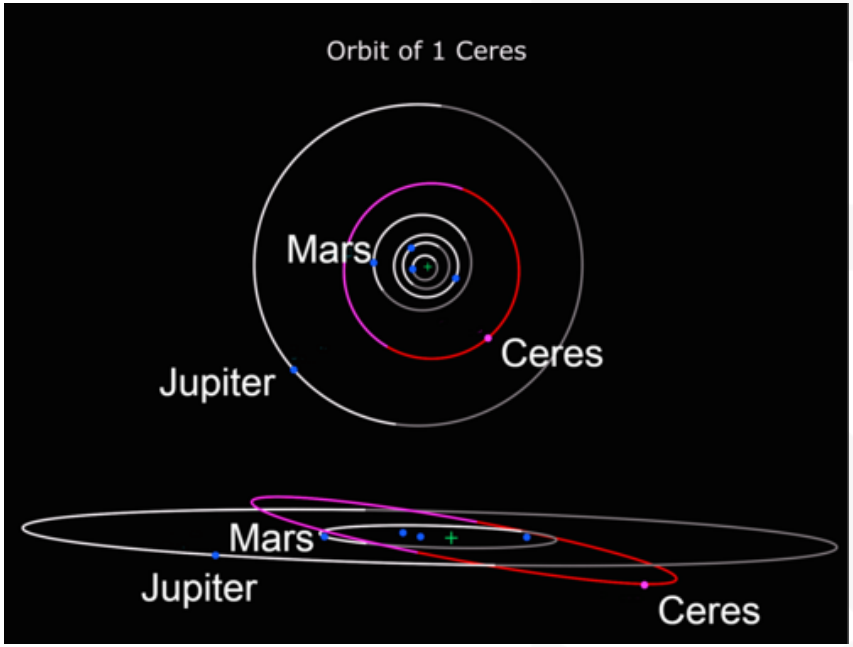
The Secret History of Least Squares

The Celestial Mystery (1801)

- Crisis: Astronomer Piazzi discovered asteroid **Ceres** but lost it after 40 days
- Challenge: Scientists couldn't predict its return with fragmentary data

Gauss' Secret Weapon

- 24-year-old Gauss used his unpublished method: least squares
- Predicted Ceres' position with just 3 observations
- Verified: Rediscovered Dec 31, 1801
- But kept it secret for 9 years!



The Priority Dispute

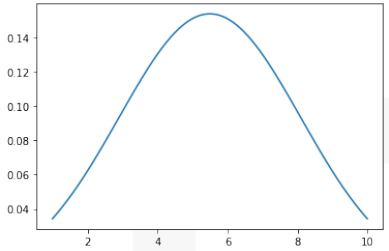
Legendre (1806)

- Published first: **Nouvelles méthodes...**
- Defined: $\min \sum \varepsilon_i^2$
- Angry letter to Gauss: **"You've taken my glory!"**

Gauss (1809)

- Revealed method in **Theoria Motus**
- Linked to normal distribution
- Claimed: **"Used since 1795"**

Historical Resolution



- **Legendre**: First publication (1806)
- **Gauss**: First application (1801) + Statistical foundation
- **Term**: "Gaussian distribution" honors Gauss' contribution

The Priority Dispute

Why did Gauss wait?

- **Perfectionism**: Needed to develop normal distribution theory
- **Strategic**: Wanted ironclad proof before release
- **Historical**: Newton had similar unpublished work in 1670s

Simplified Ceres Orbit Problem

Historical context: Gauss used only 3 observations to predict Ceres' position

Simplified data (fictional)

- Observation time (days): $t = [0, 20, 40]^T$
- Celestial longitude (degrees): $\mathbf{y} = [30.5, 31.8, 33.1]^T$

Assumption: Linear motion model (for simplicity)

$$\text{Longitude} = w_0 + w_1 \cdot t$$

Design matrix X:

$$\mathbf{X} = \begin{bmatrix} 1 & 0 \\ 1 & 20 \\ 1 & 40 \end{bmatrix}$$

Matrix Calculations Step-by-Step

Step 1: Compute $\mathbf{X}^T\mathbf{X}$

$$\mathbf{X}^T\mathbf{X} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 20 & 40 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 20 \\ 1 & 40 \end{bmatrix} = \begin{bmatrix} 3 & 60 \\ 60 & 2000 \end{bmatrix}$$

Step 2: Compute $(\mathbf{X}^T\mathbf{X})^{-1}$

$$\det = (3)(2000) - (60)(60) = 6000 - 3600 = 2400$$

$$(\mathbf{X}^T\mathbf{X})^{-1} = \frac{1}{2400} \begin{bmatrix} 2000 & -60 \\ -60 & 3 \end{bmatrix} = \begin{bmatrix} 5/6 & -1/40 \\ -1/40 & 1/800 \end{bmatrix}$$

Step 3: Compute $\mathbf{X}^T\mathbf{y}$

$$\mathbf{X}^T\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 20 & 40 \end{bmatrix} \begin{bmatrix} 30.5 \\ 31.8 \\ 33.1 \end{bmatrix} = \begin{bmatrix} 95.4 \\ 1844 \end{bmatrix}$$

Solution and Prediction

Step 4: Compute weights $\mathbf{w}^* = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$

$$\begin{aligned}
\mathbf{w}^* &= \begin{bmatrix} 5/6 & -1/40 \\ -1/40 & 1/800 \end{bmatrix} \begin{bmatrix} 95.4 \\ 1844 \end{bmatrix} \\
&= \begin{bmatrix} (5/6)(95.4) + (-1/40)(1844) \\ (-1/40)(95.4) + (1/800)(1844) \end{bmatrix} \\
&= \begin{bmatrix} 79.5 - 46.1 \\ -2.385 + 2.305 \end{bmatrix} = \begin{bmatrix} 30.4 \\ -0.08 \end{bmatrix}
\end{aligned}$$

Orbit model

$$\widehat{\text{longitude}} = 30.4 - 0.08 \cdot t$$

Prediction for t = 60 days:

$$\hat{y}(60) = 30.4 - 0.08 \times 60 = 30.4 - 4.8 = 25.6^\circ$$

Historical Validation

Actual observation on day 60: 25.7° (Gauss' prediction error: 0.1°)

- Predicted position: 25.6°
- Actual position: 25.7°
- Error: 0.1° (arcminutes)

Why was Gauss' real prediction better?

- Used elliptical orbit model (nonlinear)
- Incorporated more complex physics
- But linear approximation gave remarkable accuracy

Modern recreation

Gauss' actual 1801 prediction error: 0.25° (vs competitors' 10-20° errors)

Scientific Legacy

Impact on Modern Science:

- Astronomy: Calculating orbits of Neptune (1846) and Pluto (1930)
- Geodesy: Surveying the Earth's shape
- Statistics: Foundation of regression analysis
- Machine Learning: Core of linear models (Regularization: LASSO, Ridge, etc)

Reflection Question

Would science have advanced faster if Gauss published immediately?

Probably not - the statistical foundation was essential for modern applications